

ACC SCIENTIFIC STATEMENT

Quantitative Coronary Plaque Analysis in Clinical Practice: 2025 ACC Scientific Statement

A Report of the American College of Cardiology

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ABSTRACT

Technological advances in coronary computed tomography angiography and artificial intelligence have resulted in an increasing capability to analyze and quantify information about atherosclerotic coronary plaque through noninvasive imaging. This has paved the way for a growing number of U.S. Food and Drug Administration–cleared products that can perform quantitative coronary plaque analysis (QCPA). To date, research has focused on the accuracy, prognostic value, and decision-making impact of QCPA, but there is no current consensus on its appropriate use in clinical practice. To address this gap, the American College of Cardiology convened a panel of experts for a 1-day symposium to discuss key questions related to the use of QCPA in clinical practice and develop consensus recommendations to guide cardiovascular clinicians and imagers on the use of QCPA. This scientific statement provides guidance on clinical indications (including possible uses of QCPA in serial imaging), methods for interpretation and reporting, and standardization. Future research directions are also addressed, including both the collection of ongoing registry data and potential for incorporation of QCPA into outcomes trials.

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INTRODUCTION

The American College of Cardiology (ACC) has a long history of developing documents to complement clinical practice guidelines. Among these documents, scientific statements represent a novel approach to inform clinicians about areas where the scientific evidence is new and evolving or where data are more limited.

Recent advances in coronary computed tomography angiography (CTA) and artificial intelligence (AI) have resulted in new capabilities to rapidly analyze coronary CTA data and quantify atherosclerotic coronary plaque. This has led to widespread interest among clinicians looking to adopt this technology into clinical practice, as well as by industry seeking to incorporate these technologies into their offerings. Multiple U.S. Food and Drug Administration (FDA)-cleared commercial products are available for quantitative coronary plaque analysis (QCPA). Although QCPA provides an opportunity to accurately detect the presence, amount, and type of coronary plaque, there is insufficient evidence that QCPA improves patient management decisions and outcomes. Initial published data on QCPA have focused on measures of accuracy, prognostic value, and impact on therapeutic decision making.^{1,2} However, there is no consensus on when or how to use and interpret QCPA in clinical practice, as well as a paucity of data on the utility of serial plaque analysis.

As this is a new and rapidly evolving technology with uncertainty on how to best use QCPA in clinical practice, this ACC scientific statement aims to provide recommendations to cardiologists who use QCPA in clinical practice and to cardiovascular imagers who may interpret and report this information, while also outlining key evidence gaps for future investigation.

DEFINITIONS AND CLASSIFICATIONS

Coronary plaque: The traditional definition of plaque on coronary CTA is the presence of abnormal tissue structures $\geq 1 \text{ mm}^2$ within the coronary arterial wall with or without extension into the adjacent coronary artery lumen, identified in at least 2 independent planes, that can be distinguished from the surrounding tissues (epicardial fat) and the lumen. When performing QCPA, subvisual measures of plaque can be detected that are not identified or reported by the coronary CTA imager. Because the significance of such measures is uncertain, we propose that coronary plaque continue to be based on the presence of visually identified coronary plaque (Figure 1).

High-risk plaque: Plaque which has characteristics associated with a higher risk of incident adverse coronary events. To increase specificity over prior definitions, the authors propose a working definition of high-risk plaque (HRP) to be a plaque with mild-moderate stenosis that is positively remodeled (ie, remodeling index >1.2) and which has a component of lipid-rich plaque ($<70 \text{ HU}$). However, the authors acknowledge that technological advances (eg, high-resolution imaging, radiomics-based analysis) may prompt further refining of the definition of HRP.

Quantitative coronary plaque analysis: Analysis of coronary CTA data that identifies and quantifies (in mm^3) the overall volume of coronary plaque. QCPA may use various techniques ranging from manual segmentation to AI to distinguish between the coronary lumen and vessel wall. Total plaque volume (TPV) must be defined in relation to the number of included segments. QCPA must be defined in terms of both the degree of positive remodeling as well as integrated data on stenosis severity, the extent of obstructive coronary artery disease (CAD), and the degree of ischemia, if such information is available. Definitions of plaque quantification should also be expanded to identify the “worst lesion” and patterns (eg, “extensive” or “diffuse”) of plaque.

Plaque composition: Traditionally, plaque composition was based on broad categories: noncalcified and calcified plaque, with low-attenuation plaque typically defined as the lowest-attenuation noncalcified plaque component. Further categorization based on HU thresholds has also been used to distinguish fibrofatty from fibrous plaque.³ Plaque composition

Abbreviations

Abbreviation	Meaning/Phrase
AI	artificial intelligence
CAD	coronary artery disease
CAD-RADS	Coronary Artery Disease Reporting and Data System
CPV	calcified plaque volume
CT	computed tomography
CTA	computed tomography angiography
HRP	high-risk plaque
kVp	kilovoltage peak
MI	myocardial infarction
NCPV	noncalcified plaque volume
QCPA	quantitative coronary plaque analysis
TPV	total plaque volume

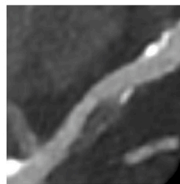
FIGURE 1 Definitions of Coronary Plaque

What is coronary plaque?

Accumulation of lipid, inflammatory cells, fibrous tissue, and at times calcium within the wall of a coronary artery.

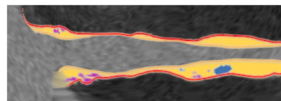
Visual plaque

Plaque that is identified visually by a physician with expertise in plaque imaging and is distinct from the lumen or normal vessel wall. On CCTA, it can be visualized on multiple planes / phases of the cardiac cycle.



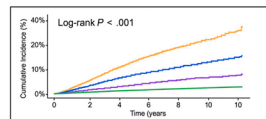
Quantitative plaque

Plaque that is identified and measured by AI or another algorithm which aims to distinguish between the normal vessel wall and the vessel lumen.



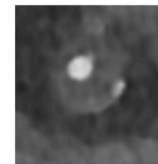
Significant plaque

Plaque which has prognostic significance (i.e. higher risk than having no plaque); defining this requires cohorts with long term follow up data.



High risk plaque

Plaque characteristics which are associated with a higher risk of adverse coronary events related to the plaque of interest.



AI = artificial intelligence; CTA = computed tomography angiography.

has several important limitations including a lack of standardization for imaging and reconstruction algorithms that increase measurement variability. Improvements in reducing calcium blooming artifact, partial volume averaging, and separation of lumen and vessel wall boundaries are necessary to create reliable quantitative measures of plaque subcategories.

Document creation process

The ACC assembled key experts in the field, including experts in coronary CTA, preventive cardiology, radiology, clinical trials, and statistical methods. This group was invited to a 1-day symposium on QCPA, held at the ACC Heart House on September 8, 2025. Also included in this meeting were representatives from the FDA and industry. The planning committee prepared questions in advance of the symposium to facilitate focused discussion. After a structured discussion on each question, attendees were then asked to vote, with those attending as representatives of industry partners or the FDA abstaining from these votes. The meeting was divided into several sections, and at the end of each section after sufficient time for discussion, a vote on key recommendations was held. Voting was performed electronically as to keep responses confidential. Consensus was agreed on if at

least two-thirds of the attendees supported a key recommendation, and was achieved on all recommendations, as displayed in [Table 1](#).

After the symposium, a Writing Committee was formed. Writing assignments were configured according to the committee members' areas of expertise. The Writing Committee's work was supported only by the ACC without any commercial input. Writing Committee members were all unpaid volunteers. In accordance with ACC's policy on Relationships With Industry and Other Entities, relevant disclosures for the Writing Committee and comprehensive disclosures for external peer reviewers can be found in [Appendixes 1 and 2](#).

Background

Evolution of coronary CTA technologies over time

Numerous coronary CTA technological advancements over the past several decades, both in terms of software and scanner hardware, have enabled high-resolution 3-dimensional imaging of coronary anatomy, and other cardiac findings.^{4,5} Advances in multidetector technology, temporal and spatial resolution, iterative reconstruction algorithms, and recently, photon counting detectors⁵ allow imaging of the entire spectrum of coronary atherosclerosis in

TABLE 1 Summary of Consensus Recommendations**Clinical indications for QCPA**

- Among patients who have visual evidence of plaque on coronary CTA, adding QCPA may be useful for enhancing risk assessment and guiding the initiation or intensification of preventive therapies.

Interpretation and reporting

- The interpreting physician must perform confirmatory review on output provided by the QCPA vendor.
- QCPA should only be performed and reported when there is clinical certainty about the presence of coronary plaque (ie, plaque is present on visual read).

The physician QCPA report should contain the following elements**Required:**

1. Quantitative QCPA results: TPV, NCPV, and CPV.
2. Qualitative QCPA results: Qualitative statement on the overall amount of plaque.
3. Total evaluated length of coronary tree, and segments excluded, if any.
4. Comparison to prior scan, if available.

Optional:

1. Vessel- and lesion-level information.
2. Percentage of atheroma volume.
3. Percentage of calcified vs noncalcified; indexing based on segments analyzed if segments were excluded.
4. Percentile score comparing plaque volumes with age- and sex-matched control subjects.
5. HRP characteristics (may be reported in main coronary CTA report), including volume of low-attenuation plaque.
6. Treatment recommendations: encourage integration with CAD-RADS 2.0.
7. Stenosis severity based on QCPA analysis.
8. Degree of ischemia, if performed.
9. Representative images.

Vendor report (PDF) should include the following elements

- Name of vendor/software.
- Image/analysis quality assessment.
- Scanner/kVp.
- Statement on reproducibility data of the QCPA software.
- Quality assurance/quality control performed, including vessel diameter threshold below which plaque data cannot be provided.
- Segments excluded from analysis, if any.

Standardization of QCPA

- Each QCPA algorithm should have published validation against invasive imaging reference data, histology reference data, or both.
- Each vendor should set and report a threshold that represents a meaningful amount of total coronary plaque volume based on technical and clinical validation.
- In the near future, creation of a standard data set should be prioritized, representative of a diverse population, that can be used by different vendors for validation and standardization purposes. The data set will need to be refreshed as technology evolves.

Use of QCPA on serial imaging

- Despite biologic plausibility, the clinical benefit of serial QCPA has not been established, and it is unknown whether serial scans provide data that are incremental to the information derived from the baseline coronary CTA.
- Serial coronary QCPA may be reasonable in select clinical scenarios, if this information will result in a meaningful change to patient management.
- If repeat coronary CTA is obtained for clinical indication(s), QCPA can be performed for purposes of serial comparison provided that the same coronary segments are analyzed on both scans.
- Serial QCPA should use a minimum follow-up interval of at least 2-5 years. Whenever possible, the follow-up scan should use the same scanner, kVp, and reconstruction parameters as the initial scan.
- Serial plaque changes should be reported using these measures: TPV in mm³ and NCPV in mm³; the report should include serial changes in these measures for each vessel and in aggregate.
- The Writing Committee recognized that it is important to evaluate changes in plaque composition, and not just TPV. In addition, any changes in QCPA should also be concordant with visual evaluation of the 2 scans.
- Although a lack of certainty exists on a clinically significant threshold of change in plaque volume, plaque progression may be considered if either of the 2 criteria are identified over serial scans:
 - TPV ≥ 10 -20 mm³/year.
 - NCPV ≥ 5 -10 mm³/year.
- When reporting such changes, the follow-up duration should be reported.
- When a dominant lesion or segment of concern is observed, initial assessment of serial QCPA should include changes in category of stenosis as well as lesion-specific plaque volume changes.
- When reporting serial QCPA, the reader should ensure that the scans are comparable, including image quality and segments analyzed or excluded.

Future studies

- Future studies are needed on prognostic, diagnostic, and therapeutic implications of minimal plaque. The clinical significance of small amounts of plaque defined by QCPA that is not visualized by the reader (ie, <20 mm³) remains unknown at this time.
- Vendors should strive to develop standardized QCPA algorithms that account for differences in kVp, detector type, and reconstruction parameters when reporting QCPA results.

CAD-RADS = Coronary Artery Disease Reporting and Data System; CPV = calcified plaque volume; CTA = computed tomography angiography; HRP = high-risk plaque; kVp = kilovoltage peak; NCPV = noncalcified plaque volume; QCPA = quantitative coronary plaque analysis; TPV = total plaque volume.

epicardial coronary arteries. At the same time, improvements focused on reducing radiation exposure further supported the safety of computed tomography (CT) imaging. Beyond anatomic imaging, physiologic measures of CT-based myocardial perfusion and fractional flow reserve also expanded the capabilities of coronary CTA.

The aforementioned technical advances led to the growing use of coronary CTA for the detection of nonobstructive and obstructive CAD among symptomatic patients with stable or acute chest pain.⁶ At the same time, growing data highlighted the prognostic value of plaque burden, especially among individuals who do not have obstructive CAD.⁷⁻⁹ As a result, it is now recommended that all coronary CTA reports provide a qualitative or semiquantitative estimate of the overall amount of plaque.¹⁰ QCPA initially relied on semiautomated segmentation of lumen and vessel wall boundaries, and calculation of plaque volume, although these techniques were time-consuming and were initially only performed in research. However, recent innovations using AI-assisted automated image segmentation have markedly facilitated rapid quantification of coronary plaque. As a result, volumetric measurements of coronary plaque can be performed throughout most of the coronary tree in a short time.¹¹ This focus beyond coronary stenosis provides significant opportunities for personalized preventive care, including the potential development of serial tracking of progressive atherosclerosis.

High-risk plaque

Invasive studies show that a bulky, lipid-rich lesion predicts a high risk for acute coronary syndrome. On coronary CTA, the most robust HRP features are positive remodeling and low-attenuation plaque, but spotty calcification and napkin ring sign have also been associated with higher risk. Although these visual measures have been used prior to the availability of QCPA, it is likely that future techniques—including QCPA, AI-based plaque phenotyping, measures of vascular inflammation, and radiomics—will improve the ability to identify HRP and offer improved specificity. As opposed to current binary based assessment, largely based on visual assessment, future methods would benefit from a more quantitative and reproducible approach that can be used to identify plaque which has different levels of risk. A challenge in this field is that even invasive measures of HRP have been unable to offer sufficiently high specificity. It also remains unclear whether findings of HRP will offer improved risk assessment after accounting for QCPA-based measures of plaque burden. Evidence is

required to robustly define a QCPA lesion phenotype predictive of acute coronary syndromes, including multiple risk-based plaque and arterial measures. Because vessels with severe stenosis are already at high risk for events, the added value of HRP characteristics is likely greater for vessels that do not have severe stenosis.

Validation and accuracy of QCPA

The clinical gold standard for plaque measurements has been invasive imaging largely using intravascular ultrasound or optical coherence tomography. Optical coherence tomography has superior spatial resolution but has limited depth of measurement to ~1 mm to 2 mm.¹² As such, studies comparing QCPA with intravascular imaging have largely used intravascular ultrasound as the reference standard.^{11,13-15} In general, measures of TPV from intravascular ultrasound have been strongly correlated with QCPA ($r > 0.9$) but often in small data sets. Reduced correlations have been reported of measures of low-attenuation plaque ($r \sim 0.3$)¹³ and may be related to the small size of this plaque subgroup.³ High interobserver discordance has also been reported.¹⁶ Furthermore, demonstrating QCPA to have reasonably high accuracy in higher risk, invasively imaged patients does not assure similarly high correlation in lower-risk patients referred to coronary CTA. Model performance is optimized for derivation data, and accuracy is only acceptable with extensive validation, especially across lower-risk patients who undergo coronary CTA. A further limitation of QCPA is that it may be difficult to distinguish small areas of noncalcified plaque from image noise or motion artifact (with edge blurring) from a healthy vessel wall. As a result, overestimation of disease prevalence in lower-risk patients may occur, including subvisual disease that is not visually apparent.²

QCPA measures and major CAD events

A critical step in defining QCPA accuracy is its ability to effectively risk-stratify patients with results yielding graded increases in event risk with increasing volumetric measures. High accuracy in risk prediction is essential for establishing the utility of QCPA, and ideally it should improve risk discrimination. This expectation should hold especially true for hard CAD events, such as myocardial infarction (MI) and CAD-related death. To date, much of the available prognostic evidence would be impacted by reduced image quality from older scanners, affecting up to 1 in 5 patients.¹⁷ Excluding small epicardial segments (from proximal to distal) can also affect accuracy, especially in women and in patients with smaller lumen volumes.³ Consequently, apparent

prognostic performance may vary across cohorts because of differences in image quality and segment inclusion criteria.

Beyond these epidemiologic metrics, understanding a patient's risk is essential to guide the initiation and intensification of preventive care, similar to targeted treatment based on a coronary artery calcium or atherosclerotic cardiovascular disease risk score. Thus, a critical question is: what QCPA measures effectively risk-stratify patients? In brief, multiple series have evaluated the association of QCPA measures with adverse cardiovascular events.^{11,18-24} From the SCOT-HEART (Scottish Computed Tomography of the Heart) trial (N = 1,769), Williams et al²⁵ reported that low-attenuation plaque had the highest hazard for MI even when controlling for the atherosclerotic cardiovascular disease risk score, coronary artery calcium score, and stenosis severity. A low-attenuation plaque burden of >4% was associated with an HR of 4.7 ($P < 0.001$).²⁵

In the EMERALD II (Exploring the Mechanism of Plaque Rupture in ACS using coronary CTA and Computational Fluid Dynamics II) study, machine learning models identified TPV, low-attenuation plaque volume, and HRP features as key determinants of culprit-lesion acute coronary syndrome risk.²¹ Early studies such as EMERALD II, however, may have been optimized to separate higher vs lower plaque volumes, potentially overstating discrimination. Much of this early evidence provides preliminary insight as to the prognostic value of QCPA; however, evidence is unavailable to predict type of MI—spontaneous vs procedural. Preliminary approaches incorporating myocardium-at-risk estimation (eg, using the Voronoi methods)²⁶ may further guide revascularization decisions. Evidence suggests that plaque erosion is more frequent among patients with non-ST-segment elevation MI than in ST-segment elevation MI, yet the descriptive phenotype has not been explored using coronary CTA. Results may also be insightful as to higher-risk findings associated with type 2 MI, spontaneous coronary artery dissection, or for other forms of MI with nonobstructive CAD.

Different studies have used varying thresholds of TPV to define risk. Lin et al¹¹ showed that $TPV \geq 238.5 \text{ mm}^3$ was associated with an HR of 5.4 ($P = 0.0042$), independent of risk factors and stenosis severity.¹¹ Other studies have used a cutoff value of $>179 \text{ mm}^3$.²⁷ Recent studies have sought to define a threshold of QCPA below which the risk of adverse cardiovascular events is very low.²⁸ These findings highlight variability in risk thresholds and underscore the need for additional evidence to define high-

risk subgroups across varied populations at varied risk of major CAD events.

Interpretation is further complicated by endpoint heterogeneity and limited follow-up. Many prognostic studies use composite endpoints (all-cause death and coronary revascularization) and relatively short follow-up (2 to 3 years), which can influence model performance compared with analyses focused on acute coronary syndrome both in the near- and longer-term follow-up. A primary focus of future studies assessing the association of QCPA with prognosis should be to establish the association of QCPA with major cardiac events.

Advantages and disadvantages of QCPA

As opposed to semiquantitative and visual estimates of plaque burden, QCPA provides a more objective and reproducible measure of calcified and noncalcified plaque burden (Table 2). The quantification of disease may better define disease severity, risk factor treatment targets, and choice of therapies, and potentially allow for monitoring of progression. Disease quantification in other fields of medicine is commonplace, allowing imaging to be used as a biomarker (eg, in oncology). With recent advances in AI, QCPA can be performed rapidly and at large scale. By capturing data on the overall amount of plaque and plaque composition, QCPA has the potential to improve prognostic accuracy beyond current measures such as coronary artery calcium score or stenosis severity.¹¹ However, issues of collinearity remain that require careful consideration in the modeling process and, as such, more data are needed. Initial data from the DECODE (Decisions for Treating Coronary Disease) study and the DECIDE (AI Derived Plaque Quantification: CCTA and AI-QCPA for Determining Effective CAD Management)²⁹ registry have shown that data from QCPA may improve patient management decisions.

Despite these advantages, QCPA is currently limited by its cost, lack of wider availability, and unclear clinical benefit over current clinical measures of CAD severity (eg, stenosis, coronary artery calcium score, ischemia). Multiple vendors and software algorithms also can be used with no standardization across different platforms. Accordingly, uncertainty exists on how to use QCPA in patient management decisions. For example, more data are needed to determine how TPV and noncalcified plaque volume (NCPV) categories relate to patient risk and how to use this information when selecting among different treatment options and defining treatment goals. Another concern is the potential that QCPA can identify plaque that is not real (false positive),

leading to unnecessary concern and overtreatment. From 1 recent report, use of QCPA resulted in a 91% and 85% prevalence of plaque in men and women, respectively, and was decidedly higher when compared with visual interpretation (63% in men, 34% in women).³⁰ This may be related to limitations in spatial resolution of current energy integrating detector CT, making it challenging to accurately differentiate healthy coronary arterial wall tissue from early noncalcified plaque, an issue that may be significantly improved with photon-counting coronary CTA. Although less likely, a potential also exists for missing real plaque (false negative) and thus underutilizing preventive therapies.² In addition, providing information on QCPA—especially without clear guidance and data regarding its prognostic value—could result in unnecessary downstream testing due to uncertainty about the test result. Last, the improved resolution of coronary CTA will continue to expand our imaging capabilities. As a result, rethinking the definitions of plaque, HRP, levels of plaque thresholds to define significant risk, and even how to measure plaque itself will be important.

Regulatory perspective

From a regulatory standpoint, QCPA software faces unique challenges. The FDA has emphasized the need for clear evidence supporting both analytic validity (accuracy and reproducibility of plaque quantification) and clinical validity (predictive value for outcomes). A particular hurdle arises as algorithms evolve: FDA guidance requires manufacturers to submit new validations or regulatory filings with each significant software update, which can slow innovation. In prior discussions, the FDA has suggested adaptive frameworks for handling future algorithm versions, to allow for more nimble model training and refinement, but these remain an ongoing challenge for the field. Developing standardized approaches for validating evolving algorithms, perhaps through modular approvals, collaborative registries, or using central standardized, validated plaque reference data sets, represents both a need and an opportunity to streamline the regulatory pathway while ensuring patient safety and data integrity.

One example of the FDA's effort to streamline the evaluation process for iterations of approved devices is the guidance document issued in August 2025 entitled "Marketing Submission Recommendations for a Predetermined Change Control Plan for Artificial Intelligence-Enabled Device Software Functions."³¹ The FDA looks forward to further efforts by a consortium of academia and industry to standardize classification systems for plaque identified by QCPA

TABLE 2 Advantages and Limitations of QCPA

Advantages	Limitations
More objective (vs visual)	Potential to overcall plaque
More reproducible (vs visual)	Small volumes of plaque may not be prognostically meaningful
Strong correlation of TPV with invasive measures of plaque ^a	Lower correlation for certain plaque subtypes (eg, low-attenuation plaque)
Can improve risk assessment	Limited prognostic and outcomes data
May enhance patient management by applying plaque-based thresholds for treatment initiation/intensification	Further data needed on how to incorporate QCPA in patient management decisions
	Cost/added time for interpreting physician
	Requires excellent image quality
	Lack of standardization across different vendors/scanners

^aFor some vendors. May depend on the vendor.

QCPA = quantitative coronary plaque analysis; TPV = total plaque volume.

as well as to efforts to harmonize the output from various platforms and vendors. Advances in these areas will accelerate the process of evaluating both new devices/software and the process of evaluating new applications of existing devices.

Clinical indications for QCPA

The group discussed and agreed on the following indication for QCPA: among patients who have visual evidence of plaque on coronary CTA, adding QCPA can be useful for enhancing risk assessment and deciding on the intensity of preventive therapies (Table 1). This indication reflects that at the current time, QCPA should be limited to patients who have visual evidence of plaque on coronary CTA (Figure 1).

The Writing Committee concluded that more research is needed about the added value of QCPA when no visual plaque (ie, Coronary Artery Disease Reporting and Data System [CAD-RADS] 0) is observed. A specific concern is whether a small amount of plaque that is identified via AI techniques is prognostically relevant, and whether it should impact patient management. In addition, it is unclear whether plaque that is identified by QCPA alone represents true plaque, or whether it could be related to image noise. In other words, it was felt that the specificity of identifying prognostically relevant plaque in patients with no visual plaque (ie, CAD-RADS 0) is reduced.

The potential benefit of QCPA is enhancing risk assessment and guiding the intensity of preventive therapies. Nevertheless, the Writing Committee recognizes that there is insufficient evidence showing that this approach improves patient outcomes. This indication also reflects that the value of QCPA may extend to both symptomatic and asymptomatic

individuals, although the committee recognizes that at the present time coronary CTA is not recommended in asymptomatic individuals.

Although coronary CTA and QCPA are commonly performed in individuals who do not have known CAD, further data are needed to better define the potential role of QCPA among individuals with severe CAD, including those with prior coronary revascularization. In addition, possible advantages may exist to guide revascularization based on QCPA, particularly when integrated with fractional flow reserve-CT and measures of myocardium at risk, but these are yet to be explored.

Consensus recommendation: clinical indications for QCPA

- Among patients who have visual evidence of plaque on coronary CTA, adding QCPA can be useful for enhancing risk assessment and deciding on the intensity of preventive therapies.

Interpreting QCPA findings

The interpretation of QCPA should be performed by a physician who has acquired the necessary knowledge and skills in this area.³² The interpreting physician must evaluate and interpret the raw coronary CTA data, including assessing for adequate image quality and the presence of any artifacts (eg, misregistration, motion, noise), and identify all lesions that have plaque. Subsequently, the interpreting physician should perform a confirmatory review on the output provided by the QCPA software, identify if any part of the coronary artery tree was excluded, and resolve any potential discrepancies.

Standards for a QCPA report

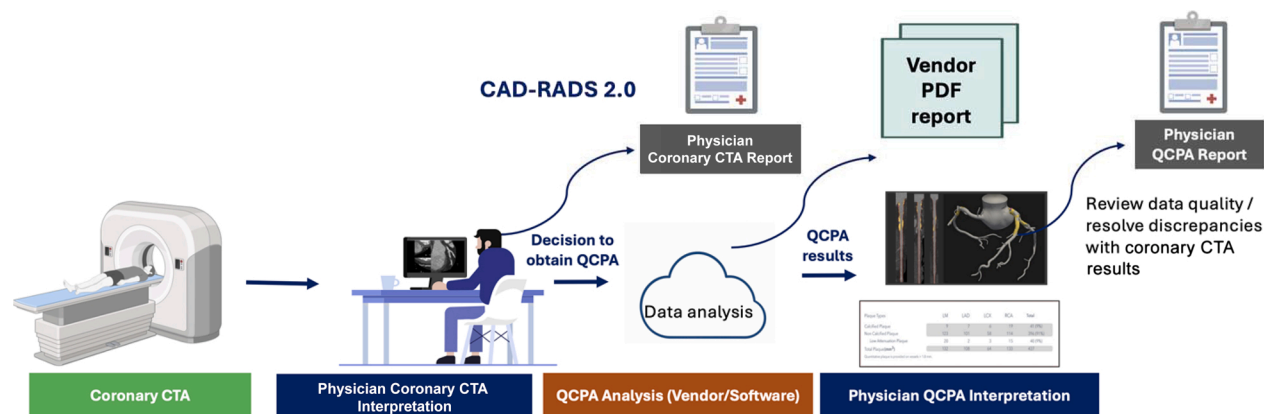
When reporting the outcome of QCPA, the Writing Committee focused on a few principles and assumptions. First, in addition to the physician QCPA report, a vendor report should also be made available. To minimize redundancy, and in recognition of the considerable extra work that interpreting physicians face when incorporating QCPA into clinical practice, most elements that are present in the vendor report do not need to be repeated in the physician report. Furthermore, the Writing Committee divided the physician report into elements that are required with every report vs those that are often desirable but at this time can remain optional.

With respect to physician reports, the Writing Committee recognizes that various workflows are available for reporting coronary CTA and QCPA findings. Ideally, a single report that includes and integrates both coronary CTA (ie, predominantly

qualitative data based on physician interpretation, which may or may not be aided by AI) and QCPA (ie, detailed quantitative data on amount and type of plaque) data would be preferable. However, because of workflow logistics, currently it is common for the coronary CTA to be reported first and then for the QCPA to be reported after this information becomes available (**Figure 2**).

The coronary CTA report should follow current reporting guidelines,¹⁰ including a qualitative assessment of stenosis severity. Physician QCPA reports should have these 4 elements:

1. Quantitative QCPA results: TPV, NCPV, and calcified plaque volume (CPV).
2. Qualitative QCPA results: Based on the TPV for the patient, the report should have a qualitative statement on the overall amount of plaque. For example, "Overall, there is a mild/moderate/severe/extensive amount of coronary plaque." As part of this statement, the interpreting physician can reconcile any differences with the main coronary CTA report (eg, the plaque [P] score category used in CAD-RADS 2.0), if this was reported prior to QCPA. A qualitative statement regarding the number of vessels with plaque and the presence of nonobstructive stenosis, obstructive stenosis, or both can also be reported in this section. For example, "Overall, there is a moderate amount of nonobstructive plaque involving 3 vessels." **Table 3** provides 2 QCPA classification systems that have been used to describe the overall amount of plaque. It is anticipated that the exact thresholds used to define the overall amount of plaque may evolve over time.
3. Total evaluated length of coronary tree, and segments excluded, if any: If any portion of the coronary artery tree was excluded because of image quality or artifacts, this information should be conveyed in the report. Alternatively, the interpreting physician may choose to report to what extent the coronary vessel tree was analyzed, if the analysis does not represent the entire coronary artery tree. This recommendation does not pertain to small or distal vessels under the threshold used for analysis (eg, <2 mm). The Writing Committee recognized that reporting this information is particularly important for the purposes of future comparisons, because omitting coronary segments with plaque could lead to an erroneous finding of plaque progression if those segments are evaluated on future QCPA examinations. The Writing Committee further discussed whether normalizing plaque volume to the total coronary bed that was

FIGURE 2 Typical Workflow of Analyzing and Reporting Coronary CTA and QCPA

CAD-RADS = Coronary Artery Disease Reporting and Data System; QCPA = quantitative coronary plaque analysis; other abbreviation as in [Figure 1](#).

analyzed would be necessary or useful and agreed that this could account for differences between studies and improve some aspects of interpretation. In the absence of studies showing benefit, the committee could not reach an agreement; however, the committee agreed that the length of the coronary tree analyzed should be reported, particularly as such information may be helpful for the purposes of future comparison.

4. Comparison to prior scan, if available: When prior scans are available, physician reports should have a comparison to the prior scan, especially if QCPA was also performed on the prior scan. When comparing to prior QCPA reports, the reader should recognize that significant differences in image quality, length and number of segments of the coronary artery tree analyzed, and differences in scan performance (eg, scanner model, detector material, slice thickness, kilovoltage peak [kVp]) can significantly impact total coronary plaque volumes and composition, and should be considered before reporting a clinical assessment regarding coronary disease progression or change.

Optional data elements ([Table 4](#)) for the physician report may include vessel and plaque (lesion) level information; percent atheroma volume; percent of total plaque that is calcified vs noncalcified; indexing based on segments analyzed, if there are excluded segments; percentile scores based on standardizing the amount of plaque relative to age and sex (and in the future perhaps also racial and ethnic background); HRP characteristics (recognizing that this information may be reported in main coronary CTA report); and amount or proportion of low-attenuation plaque.

It is also optional to provide data on stenosis severity, if performed by the AI software used for QCPA. When this is provided, the interpreting physician should reconcile any differences in stenosis severity that may exist between the coronary CTA report and the QCPA report. Finally, the QCPA report can also include treatment recommendations, and when provided, these should align with and be integrated with the CAD-RADS 2.0 document or future versions of this document or future clinical practice guideline recommendations.

The vendor report should include these elements: Name of vendor/software; assessment of image quality and how it relates to the overall quality of the assessment; scanner used; tube potential (kVp); vendor statement on reproducibility and variability data of technique used; type of quality measures used as part of analysis, including threshold below

TABLE 3 Examples of QCPA Classification Systems

Example 1 ³⁰		Example 2 ³³	
Amount of Plaque	Total Plaque Volume, mm ³	Amount of Plaque	Total Plaque Volume, mm ³
Mild	1-100	Mild	1-249
Moderate	101-250		
Severe	251-750	Moderate	250-749
Extensive	>750	Severe	>750

There are currently 2 proposed systems that categorize the amount of plaque base on total coronary plaque volume from QCPA. The main differences relate to patients with lower volumes of plaque (ie, mild vs moderate). These are initial attempts at defining subgroups that are related to risk levels, and future revised/updated classification systems with additional prognostic findings will be likely.

Abbreviation as in [Table 1](#).

TABLE 4 Elements for Physician QCPA Report

Required:

1. Quantitative QCPA results: TPV, NCPV, and CPV
2. Qualitative QCPA results: qualitative statement on the overall amount of plaque
3. Segments excluded, if any
4. Comparison to prior scan, if available

Optional:

1. Vessel- and plaque-level information
2. Percentage atheroma volume
3. Percentage calcified vs noncalcified; indexing based on segments analyzed (if segments excluded)
4. Percentile score
5. HRP characteristics (may be reported in main coronary CTA report), including amount of low-attenuation plaque
6. Treatment recommendations; encourage integration with CAD-RADS 2.0
7. Stenosis severity (by AI method)
8. Ischemia
9. Representative images

Abbreviations as in [Table 1](#).

which data would not be provided; and segments excluded from analysis, if any.

Consensus recommendations: interpretation and reporting

- Interpreting physicians must perform confirmatory review on output provided by vendor.
- QCPA should only be reported when there is clinical certainty on its presence (ie, plaque is present on visual read).
- Physician QCPA report should contain the elements listed in [Table 4](#).
- Vendor report (PDF) should include these: Name of vendor/software; image/analysis quality assessment; scanner/Kvp; statement on reproducibility data of technique uses; quality assurance/quality control performed, including threshold below which data not provided; and segments excluded from analysis, if any.

Standardization of QCPA

Plaque burden vs percentiles

The burden of coronary atherosclerosis has been consistently shown to be a powerful predictor of cardiovascular risk, whether assessed by simple measures such as segment involvement or coronary calcium, or by more advanced measures such as AI-enabled plaque quantification. These measures are strongly and independently predictive of major adverse cardiac events, revascularization, and death in both near- and long-term follow-up. In contrast, premature plaque may be identified when a smaller amount of plaque is observed that develops at a young age and is thus determined to be >70th to 75th percentile. Although less predictive of near-term events, such information is valuable for identifying individuals at increased risk for plaque progression and events over the longer term (eg, lifetime risk).

Recent studies have introduced frameworks for assessing the prematurity of atherosclerosis, stratified by age and sex.³³⁻³⁵ However, no large data sets are stratified by race and ethnicity, and only sparse data are available in younger populations, underscoring the need for broader and more inclusive studies to validate and refine percentile-based approaches. Additionally, given the lack of standardization among different software for plaque quantification, related to differences in algorithm training, vessel lumen and wall segmentation, and lengths of the coronary arteries assessed, percentiles for measures of TPV and plaque composition would currently require validation across vendors or remain specific to the software from which they were derived. Implementation of standardized approaches for QCPA are fundamental to growth in coronary CTA. Excess measurement variability across software will bring about questions of reliability and accuracy on the part of imagers and referring physicians, with the possibility of QCPA viewed as unnecessary and unhelpful to patient care. This consensus group and relevant imaging societies will continue to work with vendors on standardized approaches for QCPA to set standards on acceptable differences for QCPA, especially ranges of acceptability for total and NCPV.

Requirements for technical validation of QCPA

Each QCPA algorithm should have published validation against invasive imaging reference data, histology reference data, or both. The consensus group recognizes the need for a standard data set representative of a diverse population that can be used by different vendors for validation purposes. This may allow for future validation and calibration. Such a data set will need to be expanded and updated over time as technology evolves. With respect to validation, the consensus group also recognized that each vendor should set and report a threshold of what is a meaningful detected amount of coronary plaque. Agreed-on common standards should be set across all vendors. Identifying the smallest amount of plaque that is clinically relevant should be determined based on both technical (ie, factors related to noise/variability/reproducibility) and clinical factors (ie, prognostic value; correlation with invasive imaging).

Consensus recommendations: standardization of QCPA

- Each QCPA algorithm should have published validation against invasive imaging reference data, histology reference data, or both.
- A standard data set is needed, representative of the population, that can be used by different

vendors for validation purposes. This may allow for future validation and calibration. The data set will need to be refreshed as technology evolves.

- Each vendor should set and report a threshold of what represents a meaningful amount of plaque based on technical and clinical validation.

Standards for use of QCPA in serial imaging

Although the routine use of serial coronary CTA imaging is not supported by current clinical practice guidelines, some have suggested that such information may allow imagers and clinicians to determine whether patients are responding to medical therapy (ie, stable CAD without significant progression) or whether they have significant progression of disease and require more intensive therapy. Although such information may be appealing, the ability to assess treatment response requires: 1) accurate and reproducible measurements; 2) a well-defined threshold for identifying treatment “failure” vs “success;” 3) effective treatment options that would be implemented based on this information that would not be used otherwise based on baseline coronary CTA results alone; and 4) that the strategy of assessing serial scans would be safe (ie, exposing patients to the lowest effective dose) and cost-effective.

Despite biologic plausibility, clinical benefit of serial QCPA has not been established, and it is unknown whether data from serial scans provide information that is incremental to the information derived from the baseline coronary CTA. Thus, the Writing Committee does not recommend routine use of serial QCPA in clinical care. However, serial coronary CTA may be reasonable in select clinical scenarios, if the expected information will result in a meaningful change to patient management. For example, when repeat coronary CTA is already obtained for a clinical indication, QCPA may be performed for purposes of serial comparison. With respect to timing, when serial coronary CTAs are performed for a clinical indication, meaningful differences in QCPA are unlikely to be detected at an interval of <2 years. If performed on available scans, QCPA is thus optimally performed within a 2- to 5-year window: follow-up analysis after approximately 2 years for high-risk patients (such as those who are older, have diabetes, or smoke) and ≥5 years for lower risk individuals. Reports have noted significant progression of patients with elevated lipoprotein(a) and high-sensitivity C-reactive protein as soon as 1 to 3 years.^{36,37} The optimal time interval is much longer (eg, 7 years) for lower-risk patients or those without plaque on the index scan.³⁸⁻⁴¹ For patients who have undergone coronary

revascularization, the second scan is not interpretable if intercurrent CABG is performed. Serial interpretation can be performed if intercurrent percutaneous coronary intervention is performed but must exclude the stented segment in the analysis. In general, serial coronary CTA should only be performed if it has the potential to have a meaningful impact on patient management. Thus, for patients who are already on optimal therapy, QCPA is not indicated.

The follow-up scan should use the same scanner (ie, same vendor and model), tube voltage (kVp), and reconstruction parameters as the initial scan and, ideally, is performed using the same plaque software. Matching protocols may not be ideal, such as when the index scan was of suboptimal quality or older equipment is no longer available. When newer scanners are used, serial comparison should default to qualitative changes in plaque burden, combined with QCPA for the second scan. If serial scans are performed with different kVp, artificial differences in plaque composition will be observed due to varied HU thresholds as well as differences in image quality. In such scenarios, it is not recommended to compare QCPA results from different CT scans. Simple comments on the presence of new lesions or plaque extent (ie, the number of vessels with plaque) may be possible, depending on image quality.

When evaluating for serial changes on 2 coronary CTA scans, the vendor and reader should ensure that the scans are comparable, including having similar image quality, and that the same segments are analyzed or excluded. The report should include serial changes in each vessel. In addition, when dominant lesion or segment is of concern, the assessment should include changes in category of stenosis as well as lesion-specific changes.

Changes in serial QCPA should focus on TPV (mm³) and NCPV (mm³). The report should include serial changes in these measures for each vessel and in aggregate. However, evaluating plaque composition, and not just TPV, is also important. For example, it is well known that some noncalcified plaque may convert to calcified plaque, and that such change in composition may be favorable, even if the change does not result in any change in TPV. Any changes in QCPA should also be concordant with visual evaluation of the 2 scans.

Lack of certainty on how to define plaque progression poses a challenge; the minimum amount of plaque progression that is clinically meaningful and which exceeds the observed variance. To better define such thresholds in the future would require: 1) multicenter (possibly multivendor) test-retest studies

to quantify repeatability and compute a “least significant change” ($1.96 \times \sqrt{2} \times \text{SD of repeat measurements}$) for TPV and NCPV at the patient and vessel levels; and 2) a prognostic anchoring strategy that links absolute as well relative changes in plaque (eg, ΔTPV , ΔNCPV) to changes in predicted major cardiac event risk using validated coronary CTA risk models (eg, clinical trial data or large registries using QCPA that have robust follow-up data), so that thresholds reflect both measurement precision and clinically meaningful risk differences. Moreover, harmonized acquisition and reconstruction protocols, standardized phantoms and reference data sets for cross-software calibration, and registries that pair serial QCPA with outcomes would enable robust threshold setting, improve comparability across centers, and accelerate validation of clinically meaningful change.

Although evidence is limited, plaque progression should be based on annualized thresholds and may be considered if either of these 2 criteria are identified over serial scans: 1) increase in TPV $\geq 10 \text{ mm}^3/\text{year}$; or 2) increase in NCPV $\geq 5 \text{ mm}^3/\text{year}$ to $10 \text{ mm}^3/\text{year}$. However, disease progression is not equivalent on a yearly basis and will be greater in the near term for patients with a higher baseline plaque burden. Percentage change can be informative, though less reliable with small baseline volumes. The Writing Committee recommends annualized thresholds for defining progression on serial scans.

Of the various changes in plaque composition that can be identified on coronary CTA, NCPV represents the most valid measure for follow-up, and is the one that is being used in coronary CTA-based clinical trials.^{42,43} Unlike CPV, NCPV can be modifiable and is more stable than low-attenuation plaque. Robust data show that changes in NCPV are associated with favorable changes in outcomes.^{44,45} A burgeoning role is noted for integrating mechanistic coronary CTA substudies within larger research projects and clinical trials.^{44,46} This may provide insights into the main trial results, either negative or positive, while also providing further data to help establish the role of NCPV as a valid surrogate endpoint.

Consensus recommendations: use of QCPA in serial imaging

- The Writing Committee does not recommend routine use of serial QCPA in clinical care.
- Serial coronary CTA may be reasonable in select clinical scenarios, if this information will result in a meaningful change to patient management. If serial coronary CTAs are available, QCPA can be considered with a minimum interscan follow-up

interval of at least 2 to 5 years, depending on the baseline risk. Whenever possible, the follow-up scan should use the same scanner, kVp, and reconstruction parameters as the initial scan.

- If repeat coronary CTA is obtained for clinical indication, QCPA can be performed for purposes of serial comparison.
- Serial plaque changes should be reported using these measures: TPV in mm^3 and NCPV in mm^3 ; the report should include serial changes in these measures for each vessel and in aggregate.
- A lack of certainty exists on how to define the minimal change in plaque progression. The writing group recognizes that evaluating plaque composition, and not just TPV, is important. In addition, any changes in QCPA should also be concordant with visual evaluation of the 2 scans.
- Plaque progression may be considered if either of these 2 criteria are identified over serial scans: TPV $\geq 10 \text{ mm}^3/\text{year}$ to $20 \text{ mm}^3/\text{year}$ or NCPV $\geq 5 \text{ mm}^3/\text{year}$ to $10 \text{ mm}^3/\text{year}$. When reporting such changes, the follow-up duration should be reported.
- When a dominant lesion or segment is of concern, initial assessment of serial QCPA should include changes in category of stenosis as well as lesion-specific changes (eg, positive remodeling).
- When reporting serial QCPA, the reader should ensure that the scans are comparable, including image quality and segments analyzed or excluded.
- Each QCPA algorithm should have published validation against invasive imaging reference data, histology reference data, or both.

Evidence gaps and future directions

Although future improvement in CT hardware, reconstruction parameters, and AI based data analysis will continue to improve the capabilities of QCPA, standardizing how plaque measurements are performed will be important, especially to define future common threshold of risk or to identify standards for assessing serial changes in QCPA. Various approaches can be used to achieve this including the use of phantoms or having large coronary CTA data sets with invasive correlation that can be used by vendors for improving and standardizing their analysis. As is true for most AI techniques, having a “human in the loop” who can ensure quality of the process and that QCPA results are reproducible will be important. The human may be a specialist provided by the vendor or, in the future, the imaging physician. Regardless, it is still essential for the CT imager to know how to evaluate, interpret, and report QCPA data and to be actively involved in the

review, interpretation, and communication of the QCPA findings.

Several ongoing research endeavors are underway that will help to formulate coronary CTA-guided algorithms for patient care. For QCPA to be a meaningful part of patient care, it will need to provide results that are actionable and evidence-based.

The use of HU thresholds to define plaque composition is imperfect. Additional studies should identify whether adaptive thresholds—which, unlike fixed thresholds, rely on the contrast attenuation of the lumen—are advantageous. In addition, thresholds and segmentation algorithms for plaque quantification and composition may need to be adjusted based on the type of scanner, the acquisition protocol (eg, ultra high-resolution mode using 0.2 mm spatial resolution), and the reconstruction algorithm used. Ultimately, the goal should be to reduce variability in plaque classification and improve patient outcomes. To this end, future studies are needed to determine whether providing this information to patients and clinicians improves outcomes. In addition, QCPA may help identify high-risk patients who—by virtue of the amount or type of plaque—are more likely to benefit from certain therapies. To use QCPA in this manner, coronary CTA studies embedded in cardiovascular outcomes trials would be valuable, even if only in a subset. Such data, together with large registries with long-term follow-up data, could allow researchers to identify coronary CTA-derived thresholds that can be used in selecting various treatments. Eventually, such detailed phenotyping of coronary atherosclerosis will be combined with other blood biomarkers,

lipid parameters, and genetic risk scores to personalize treatment plans. The addition of QCPA adds considerable cost to coronary CTA. Additional studies are warranted to explore the cost-benefit ratio, especially if QCPA-guided care can reduce major CAD events.

Consensus recommendations: evidence gaps and future directions

- Future studies are needed on prognostic, diagnostic, and therapeutic implications of minimal plaque. The clinical significance of small amounts of plaque defined by QCPA that is not visualized by the reader (ie, 20 mm to 30 mm) remains unknown at this time.

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REFERENCES

1. Bell JS, Weir-McCall J, Nicol E, et al. Plaque quantification from coronary computed tomography angiography in predicting cardiovascular events: a systematic review and meta-analysis. *J Cardiovasc Comput Tomogr*. 2025;19:423–432.
2. Shaw LJ, Blankstein R, Leipsic JA, et al. Clinical integration of AI-enabled plaque quantification to improve cardiovascular risk stratification. *JACC Cardiovasc Imaging*. Published online September 8, 2025. <https://doi.org/10.1016/j.jcmg.2025.09.001>
3. Bienstock S, Lin F, Blankstein R, et al. Advances in coronary computed tomographic angiographic imaging of atherosclerosis for risk stratification and preventive care. *JACC Cardiovasc Imaging*. 2023;16:1099–1115.
4. Abbara S, Shaw LJ. Past, present, and future of CTA. *Circulation*. 2024;150:87–90.
5. Shiyovich A, Singh A, Blair CV, et al. Photon-counting computed tomography in cardiac imaging. *JACC Cardiovasc Imaging*. Published online October 6, 2025. <https://doi.org/10.1016/j.jcmg.2025.07.022>
6. Gulati M, Levy PD, Mukherjee D, et al. 2021 AHA/ACC/ASE/CHEST/SAEM/SCCT/SCMR guideline for the evaluation and diagnosis of chest pain: a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *J Am Coll Cardiol*. 2021;78:e187–e285.
7. Bittencourt MS, Hultén E, Ghoshhajra B, et al. Prognostic value of nonobstructive and obstructive coronary artery disease detected by coronary computed tomography angiography to identify cardiovascular events. *Circ Cardiovasc Imaging*. 2014;7:282–291.
8. Shaw LJ, Blankstein R, Bax JJ, et al. Society of Cardiovascular Computed Tomography/North American Society of Cardiovascular Imaging – Expert consensus document on coronary CT imaging of atherosclerotic plaque. *J Cardiovasc Comput Tomogr*. 2021;15:192–217.
9. Bianchini E, Alqahtani F, Alsubai S, et al. Advanced analyses of coronary computed tomography angiography to predict future cardiac events: a meta-analysis. *JACC Cardiovasc Imaging*. 2025;18:1220–1231.
10. Cury RC, Leipsic J, Abbara S, et al. CAD-RADS™ 2.0 – 2022 Coronary Artery Disease-Reporting and Data System: an expert consensus document of the Society of Cardiovascular Computed Tomography (SCCT), the American College of Cardiology (ACC), the American College of Radiology (ACR), and the North America Society of Cardiovascular Imaging (NASCI). *JACC Cardiovasc Imaging*. 2022;15:1974–2001.
11. Lin A, Manral N, McElhinney P, et al. Deep learning-enabled coronary CT angiography for plaque and stenosis quantification and cardiac risk

prediction: an international multicentre study. *Lancet Digit Health*. 2022;4:e256-e265.

12. Nagaraja V, Kalra A, Puri R. When to use intravascular ultrasound or optical coherence tomography during percutaneous coronary intervention? *Cardiovasc Diagn Ther*. 2020;10:1429-1444.

13. Narula J, Stuckey TD, Nakazawa G, et al. Prospective deep learning-based quantitative assessment of coronary plaque by computed tomography angiography compared with intravascular ultrasound: the REVEALPLAQUE study. *Eur Heart J Cardiovasc Imaging*. 2024;25:1287-1295.

14. Ildayhid AR, Tzimas G, Peterson K, et al. Diagnostic performance of AI-enabled plaque quantification from coronary CT angiography compared with intravascular ultrasound. *Radiol Cardiothorac Imaging*. 2024;6:e230312.

15. Hakim D, Coskun AU, Maynard C, et al. Endothelial shear stress computed from coronary computed tomography angiography: a direct comparison to intravascular ultrasound. *J Cardiovasc Comput Tomogr*. 2023;17:201-210.

16. Jonas RA, Weerakoon S, Fisher R, et al. Interobserver variability among expert readers quantifying plaque volume and plaque characteristics on coronary CT angiography: a CLARIFY trial sub-study. *Clin Imaging*. 2022;91:19-25.

17. Nurmohamed NS, Min JK, Anthonopolis R, et al. Atherosclerosis quantification and cardiovascular risk: the ISCHEMIA trial. *Eur Heart J*. 2024;45:3735-3747.

18. Chang H-J, Lin FY, Lee S-E, et al. Coronary atherosclerotic precursors of acute coronary syndromes. *J Am Coll Cardiol*. 2018;71:2511-2522.

19. Mézquita AJV, Biavati F, Falk V, et al. Clinical quantitative coronary artery stenosis and coronary atherosclerosis imaging: a consensus statement from the quantitative cardiovascular imaging study group. *Nat Rev Cardiol*. 2023;20:696-714.

20. Ferencik M, Mayrhofer T, Bittner DO, et al. Use of high-risk coronary atherosclerotic plaque detection for risk stratification of patients with stable chest pain: a secondary analysis of the PROMISE randomized clinical trial. *JAMA Cardiol*. 2018;3:144-152.

21. Koo BK, Yang S, Jung JW, et al. Artificial intelligence-enabled quantitative coronary plaque and hemodynamic analysis for predicting acute coronary syndrome. *JACC Cardiovasc Imaging*. 2024;17:1062-1076.

22. Deseive S, Kupke M, Straub R, et al. Quantified coronary total plaque volume from computed tomography angiography provides superior 10-year risk stratification. *Eur Heart J Cardiovasc Imaging*. 2021;22:314-321.

23. Dundas J, Leipsic J, Fairbairn T, et al. Interaction of AI-enabled quantitative coronary plaque volumes on coronary CT angiography, FFR(CT), and clinical outcomes: a retrospective analysis of the ADVANCE registry. *Circ Cardiovasc Imaging*. 2024;17:e016143.

24. Nurmohamed NS, Bom MJ, Jukema RA, et al. AI-guided quantitative plaque staging predicts long-term cardiovascular outcomes in patients at risk for atherosclerotic CVD. *JACC Cardiovasc Imaging*. 2024;17:269-280.

25. Williams MC, Kwicinski J, Doris M, et al. Low-attenuation noncalcified plaque on coronary computed tomography angiography predicts myocardial infarction: results from the multicenter SCOT-HEART Trial (Scottish Computed Tomography of the HEART). *Circulation*. 2020;141:1452-1462.

26. van Driest FY, Bijns CM, van der Geest RJ, et al. Correlation between quantification of myocardial area at risk and ischemic burden at cardiac computed tomography. *Eur J Radiol Open*. 2022;9:100417.

27. Hell MM, Motwani M, Otaki Y, et al. Quantitative global plaque characteristics from coronary computed tomography angiography for the prediction of future cardiac mortality during long-term follow-up. *Eur Heart J Cardiovasc Imaging*. 2017;18:1331-1339.

28. Bar S, Knuuti J, Saraste A, et al. Derivation and validation of an artificial intelligence-based plaque burden safety cut-off for long-term acute coronary syndrome from coronary computed tomography angiography. *Eur Heart J Cardiovasc Imaging*. 2025;26:1163-1173.

29. Rinehart S, Blankstein R, Januzzi JL, et al. Guiding automated implementation strategies for patients with atherosclerotic plaque on coronary computed tomographic angiography. *JACC Cardiovasc Imaging*. 2025;18:1107-1115.

30. Ichikawa K, Ronen S, Bishay R, et al. Coronary plaque volume in an asymptomatic population: Miami Heart Study at Baptist Health South Florida. *JACC Cardiovasc Imaging*. Published online September 3, 2025. <https://doi.org/10.1016/j.jcmg.2025.08.001>

31. U.S. Food and Drug Administration. Guidance Document. Marketing submission recommendations for a predetermined change control plan for artificial intelligence-enabled device software functions: guidance for Food and Drug Administration staff. August 2025. Accessed November 17, 2025. <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/marketing-submission-recommendations-predetermined-change-control-plan-artificial-intelligence>

32. Baldassarre LA, Mendes LA, Blankstein R, et al. 2025 ACC/AHA/ASE/ASNC/SCCT/SCMR advanced training statement on advanced cardiovascular imaging: a report of the ACC Competency Management Committee. *J Am Coll Cardiol*. Published online November 18, 2025. <https://doi.org/10.1016/j.jacc.2025.07.036>

33. Min JK, Chang H-J, Andreini D, et al. Coronary CTA plaque volume severity stages according to invasive coronary angiography and FFR. *J Cardiovasc Comput Tomogr*. 2022;16:415-422.

34. Tzimas G, Gulsin GS, Everett RJ, et al. Age- and sex-specific nomographic CT quantitative plaque data from a large international cohort. *JACC Cardiovasc Imaging*. 2024;17:165-175.

35. Miller RJH, Manral N, Lin A, et al. Patient-specific myocardial infarction risk thresholds from AI-enabled coronary plaque analysis. *Circ Cardiovasc Imaging*. 2024;17:e016958.

36. Bienstock S, Lee SE, Lin F, et al. Systemic inflammation with high-sensitivity C-reactive protein and atherosclerotic plaque progression. *JACC Cardiovasc Imaging*. 2024;17:212-213.

37. Kaiser Y, Daghem M, Tzolos E, et al. Association of lipoprotein(a) with atherosclerotic plaque progression. *J Am Coll Cardiol*. 2022;79:223-233.

38. Budoff MJ, Young R, Lopez VA, et al. Progression of coronary calcium and incident coronary heart disease events: MESA (Multi-Ethnic Study of Atherosclerosis). *J Am Coll Cardiol*. 2013;61:1231-1239.

39. Min JK, Lin FY, Gidseg DS, et al. Determinants of coronary calcium conversion among patients with a normal coronary calcium scan: what is the "warranty period" for remaining normal? *J Am Coll Cardiol*. 2010;55:1110-1117.

40. Radford NB, DeFina LF, Barlow CE, et al. Progression of CAC score and risk of incident CVD. *JACC Cardiovasc Imaging*. 2016;9:1420-1429.

41. Dzaye O, Dardari ZA, Cainzos-Achirica M, et al. Warranty period of a calcium score of zero: comprehensive analysis from MESA. *JACC Cardiovasc Imaging*. 2021;14:990-1002.

42. McCarthy CP, Ballantyne CM, Blankstein R, et al. Rationale and design of the REMBRANDT trial: a phase 3 study to evaluate the effect of obicetrapib/ezetimibe on coronary plaque characteristics. *Am Heart J*. 2025;290:325-338.

43. Budoff MJ, Bhatt DL, Kinninger A, et al. Effect of icosapent ethyl on progression of coronary atherosclerosis in patients with elevated triglycerides on statin therapy: final results of the EVAPORATE trial. *Eur Heart J*. 2020;41:3925-3932.

44. Baskaran L, Dutta U, Xiaomeng W, et al. Accelerating drug translation: role of coronary computed tomography angiography plaque imaging candidates as surrogate biomarkers. *JACC Cardiovasc Imaging*. 2025;18:912-927.

45. Lu MT, Ribaudo H, Foldyna B, et al. Effects of pitavastatin on coronary artery disease and inflammatory biomarkers in HIV: mechanistic sub-study of the REPRIEVE randomized clinical trial. *JAMA Cardiol*. 2024;9:323-334.

46. Howden N, Branch K, Douglas P, et al. Computed tomographic angiography measures of coronary plaque in clinical trials: opportunities and considerations to accelerate drug translation. *Front Cardiovasc Med*. 2024;11:1359500.

KEY WORDS ACC Scientific Statement, cardiac imaging, coronary artery disease, coronary computed tomography, quantitative coronary plaque analysis

APPENDIX 1: AUTHOR RELATIONSHIPS WITH INDUSTRY AND OTHER ENTITIES (RELEVANT)—QUANTITATIVE CORONARY PLAQUE ANALYSIS IN CLINICAL PRACTICE: 2025 ACC SCIENTIFIC STATEMENT

Committee Member	Employment	Consultant	Speakers Bureau	Ownership/ Partnership/ Principal	Personal Research	Institutional, Organizational, or Other Financial Benefit	Expert Witness
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Ron Blankstein, <i>Vice Chair</i>	Associate Director, Cardiovascular Imaging; Director, Cardiac Computed Tomography; Co-Director, Cardiovascular Imaging Training Program; Professor of Medicine and Radiology, Harvard Medical School	■ Caristo ■ Heartflow ■ Nanox AI* ■ Siemens ■ Novartis ■ Amgen	None	None	■ Amgen† ■ Nanox AI ■ Novartis† ■ Heartflow Inc	None	None
Leslee Shaw, <i>Vice Chair</i>	Professor, Obstetrics, Gynecology and Reproductive Science; Professor, Medicine, Cardiology; Professor, Population Health Science and Policy, Mount Sinai	None	None	None	None	None	None
Maros Ferencik	Professor of Medicine, Division of Cardiovascular Medicine, Oregon Health and Sciences University	■ Cleerly* ■ Elucid ■ HeartFlow* ■ Siemens ■ Healthineers	None	■ Elucid	None	None	None
Jonathon Leipsic	Professor and Chair of Radiology; Professor of Cardiology, University of British Columbia	■ CircleCVI* ■ HeartFlow*	None	■ CircleCVI* ■ HeartFlow*	None	None	None
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Todd C. Villines	Professor of Medicine, University of Virginia	None	None	None	None	None	None

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ACC = American College of Cardiology; AHA = American Heart Association; JACC = *Journal of the American College of Cardiology*; VA = Veterans Affairs.

APPENDIX 2: PEER REVIEWER RELATIONSHIPS WITH INDUSTRY AND OTHER ENTITIES (COMPREHENSIVE)—QUANTITATIVE CORONARY PLAQUE ANALYSIS IN CLINICAL PRACTICE: 2025 ACC SCIENTIFIC STATEMENT

Reviewer	Employment	Consultant	Speakers Bureau	Ownership/ Partnership/ Principal	Personal Research	Institutional, Organizational, or Other Financial Benefit	Expert Witness
William Herzog	Physician, Office of Product Evaluation and Quality, Center for Devices and Radiologic Health, U.S. Food and Drug Administration	None	None	None	None	None	None
Purvi Parwani	Associate Professor of Medicine; Director, Women's Heart Clinic, Loma Linda University	■ Zoll	■ BMS	None	None	None	None
Monvadi Barbara Srichai-Parsia	Professor of Medicine & Radiology, Georgetown University; Vice Chief of Cardiology Division, Director of Noninvasive Cardiac Laboratory, Director of Medstar Women's Heart Health Program, MedStar Georgetown University Hospital	■ Artrya ■ AstraZeneca† ■ Clairio*	None	None	None	None	None

*Significant relationship. †No financial benefit.

ACC = American College of Cardiology.